

Discussion Week 7

Probability & Bayesian Network

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Marginal Probability (FJDT) - Review

Example from Russell & Norvig

Marginal Probability

- Can fully specify a probability space by constructing a full joint distribution
- Example: dentist
 - T: have a toothache
 - D: dental probe catches
 - C: have a cavity
- Joint distribution
 - Assigns each event (T=t, D=d, C=c) a probability
 - Probabilities sum to 1.0
- Law of total probability:

$$p(C = 1) = \sum_{t,d} P(T = t, D = d, C = 1)$$

$$= 0.008 + 0.072 + 0.012 + 0.108 = 0.20$$

- Some value of (T,D) must occur; values are disjoint
- “Marginal probability” of C; “marginalize” or “sum over” T,D
- Early actuaries wrote row & column totals in their probability table margins

T	D	C	P(T,D,C)
0	0	0	0.576
0	0	1	0.008
0	1	0	0.144
0	1	1	0.072
1	0	0	0.064
1	0	1	0.012
1	1	0	0.016
1	1	1	0.108

Marginal Probability (FJDT) - Question

4. (10 pts total, 1 pt each) Probability Rules and Independence.

Consider the following full joint distribution for Boolean variables A, B, and C:

<i>A</i>	<i>B</i>	<i>C</i>	$P(a,b,c)$
<i>t</i>	<i>t</i>	<i>t</i>	0.03
<i>t</i>	<i>t</i>	<i>f</i>	0.12
<i>t</i>	<i>f</i>	<i>t</i>	0.17
<i>t</i>	<i>f</i>	<i>f</i>	0.18
<i>f</i>	<i>t</i>	<i>t</i>	0.03
<i>f</i>	<i>t</i>	<i>f</i>	0.12
<i>f</i>	<i>f</i>	<i>t</i>	0.24
<i>f</i>	<i>f</i>	<i>f</i>	0.11

4.a. (1 pt) $P(A = f) =$ _____

4.b. (1 pt) $P(B = t) =$ _____

4.c. (1 pt) $P(B = t, C = t) =$ _____

4.d. (1 pt) $P(A = f, C = t) =$ _____

Marginal Probability (FJDT) - Solution

4. (10 pts total, 1 pt each) Probability Rules and Independence.

Consider the following full joint distribution for Boolean variables A, B, and C:

<i>A</i>	<i>B</i>	<i>C</i>	<i>P(a,b,c)</i>
<i>t</i>	<i>t</i>	<i>t</i>	0.03
<i>t</i>	<i>t</i>	<i>f</i>	0.12
<i>t</i>	<i>f</i>	<i>t</i>	0.17
<i>t</i>	<i>f</i>	<i>f</i>	0.18
<i>f</i>	<i>t</i>	<i>t</i>	0.03
<i>f</i>	<i>t</i>	<i>f</i>	0.12
<i>f</i>	<i>f</i>	<i>t</i>	0.24
<i>f</i>	<i>f</i>	<i>f</i>	0.11

<i>A</i>	<i>B</i>	<i>C</i>	<i>P(a,b,c)</i>
<i>t</i>	<i>t</i>	<i>t</i>	0.03
<i>t</i>	<i>t</i>	<i>f</i>	0.12
<i>t</i>	<i>f</i>	<i>t</i>	0.17
<i>t</i>	<i>f</i>	<i>f</i>	0.18
<i>f</i>	<i>t</i>	<i>t</i>	0.03
<i>f</i>	<i>t</i>	<i>f</i>	0.12
<i>f</i>	<i>f</i>	<i>t</i>	0.24
<i>f</i>	<i>f</i>	<i>f</i>	0.11

$P(A=f) = 0.03+0.12 + 0.24 + 0.11 = 0.50$

<i>A</i>	<i>B</i>	<i>C</i>	<i>P(a,b,c)</i>
<i>t</i>	<i>t</i>	<i>t</i>	0.03
<i>t</i>	<i>t</i>	<i>f</i>	0.12
<i>t</i>	<i>f</i>	<i>t</i>	0.17
<i>t</i>	<i>f</i>	<i>f</i>	0.18
<i>f</i>	<i>t</i>	<i>t</i>	0.03
<i>f</i>	<i>t</i>	<i>f</i>	0.12
<i>f</i>	<i>f</i>	<i>t</i>	0.24
<i>f</i>	<i>f</i>	<i>f</i>	0.11

$P(B=t) = 0.03 + 0.12 + 0.03 + 0.12 = 0.30$

4.a. (1 pt) $P(A = f) =$ 0.5

4.b. (1 pt) $P(B = t) =$ 0.3

4.c. (1 pt) $P(B = t, C = t) =$ 0.06

4.d. (1 pt) $P(A = f, C = t) =$ 0.27

<i>A</i>	<i>B</i>	<i>C</i>	<i>P(a,b,c)</i>
<i>t</i>	<i>t</i>	<i>t</i>	0.03
<i>t</i>	<i>t</i>	<i>f</i>	0.12
<i>t</i>	<i>f</i>	<i>t</i>	0.17
<i>t</i>	<i>f</i>	<i>f</i>	0.18
<i>f</i>	<i>t</i>	<i>t</i>	0.03
<i>f</i>	<i>t</i>	<i>f</i>	0.12
<i>f</i>	<i>f</i>	<i>t</i>	0.24
<i>f</i>	<i>f</i>	<i>f</i>	0.11

$P(B=t, C=t) = 0.03 + 0.03 = 0.06$

<i>A</i>	<i>B</i>	<i>C</i>	<i>P(a,b,c)</i>
<i>t</i>	<i>t</i>	<i>t</i>	0.03
<i>t</i>	<i>t</i>	<i>f</i>	0.12
<i>t</i>	<i>f</i>	<i>t</i>	0.17
<i>t</i>	<i>f</i>	<i>f</i>	0.18
<i>f</i>	<i>t</i>	<i>t</i>	0.03
<i>f</i>	<i>t</i>	<i>f</i>	0.12
<i>f</i>	<i>f</i>	<i>t</i>	0.24
<i>f</i>	<i>f</i>	<i>f</i>	0.11

$P(A=f, C=t) = 0.03 + 0.24 = 0.27$

Independence (FJDT) - Review

Independence

- X, Y independent:
 - $p(X=x, Y=y) = p(X=x) p(Y=y)$ for all x, y
 - Shorthand: $p(X, Y) = P(X) P(Y)$
 - Equivalent: $p(X|Y) = p(X)$ or $p(Y|X) = p(Y)$ (if $p(Y), p(X) > 0$)
 - Intuition: knowing X has no information about Y (or vice versa)

Independence (FJDT) - Provement

$$p(X=x, Y=y) = p(X=x) p(Y=y) \text{ for all } x, y$$

Binary Case: Only need to prove $P(A=T, B=T) = P(A=T) P(B=T)$

Suppose we have proved that $P(A=T, B=T) = P(A=T) P(B=T)$

Then, $a = (a + b)(a + c) \Rightarrow a = a^2 + (b+c)a + bc$

B \ A	T	F
T	a	b
F	c	1-a-b-c

Here we want to prove that $P(A=F, B=T) = P(A=F) P(B=T)$

$$\begin{aligned} &P(A=F) P(B=T) \\ &= (b + 1 - a - b - c) * (a + b) \\ &= (1 - a - c) * (a + b) \\ &= a + b - a^2 - ab - ac - bc \\ &= a^2 + (b+c)a + bc + b - a^2 - ab - ac - bc \\ &= b = P(A=F, B=T) \end{aligned}$$

Independence (FJDT) - Question

4. (10 pts total, 1 pt each) Probability Rules and Independence.

Consider the following full joint distribution for Boolean variables A, B, and C:

<i>A</i>	<i>B</i>	<i>C</i>	<i>P(a,b,c)</i>
<i>t</i>	<i>t</i>	<i>t</i>	<i>0.03</i>
<i>t</i>	<i>t</i>	<i>f</i>	<i>0.12</i>
<i>t</i>	<i>f</i>	<i>t</i>	<i>0.17</i>
<i>t</i>	<i>f</i>	<i>f</i>	<i>0.18</i>
<i>f</i>	<i>t</i>	<i>t</i>	<i>0.03</i>
<i>f</i>	<i>t</i>	<i>f</i>	<i>0.12</i>
<i>f</i>	<i>f</i>	<i>t</i>	<i>0.24</i>
<i>f</i>	<i>f</i>	<i>f</i>	<i>0.11</i>

4.g. (1 pt) Are A and B independent of each other? (Y=Yes, N=No): _____

4.h. (1 pt) Are B and C independent of each other? (Y=Yes, N=No): _____

Independence (FJDT) - Solution

4. (10 pts total, 1 pt each) Probability Rules and Independence.

Consider the following full joint distribution for Boolean variables A, B, and C:

A	B	C	$P(a,b,c)$
t	t	t	0.03
t	t	f	0.12
t	f	t	0.17
t	f	f	0.18
f	t	t	0.03
f	t	f	0.12
f	f	t	0.24
f	f	f	0.11

A	B	C	$P(a,b,c)$
t	t	t	0.03
t	t	f	0.12
t	f	t	0.17
t	f	f	0.18
f	t	t	0.03
f	t	f	0.12
f	f	t	0.24
f	f	f	0.11

$P(A=t, B=t) = 0.03 + 0.12 = 0.15$
 $P(A=t) = 0.03 + 0.12 + 0.17 + 0.18 = 0.5$
 $P(B=t) = 0.03 + 0.12 + 0.03 + 0.12 = 0.3$
 $P(A=t, B=t) = P(A=t) * P(B=t)$

A	B	C	$P(a,b,c)$
t	t	t	0.03
t	t	f	0.12
t	f	t	0.17
t	f	f	0.18
f	t	t	0.03
f	t	f	0.12
f	f	t	0.24
f	f	f	0.11

$P(B=t, C=t) = 0.03 + 0.03 = 0.06$
 $P(B=t) = 0.03 + 0.12 + 0.03 + 0.12 = 0.3$
 $P(C=t) = 0.03 + 0.17 + 0.03 + 0.24 = 0.47$
 $P(B=t, C=t) \neq P(B=t) * P(C=t)$

4.g. (1 pt) Are A and B independent of each other? (Y=Yes, N=No): Y

4.h. (1 pt) Are B and C independent of each other? (Y=Yes, N=No): N

Posterior Conditional (FJDT) - Review

The effect of evidence

Example from Russell & Norvig

- Example: dentist
 - T: have a toothache
 - D: dental probe catches
 - C: have a cavity

- Recall $p(C=1) = 0.20$

- Suppose we observe $D=0, T=0$?

Product rule $\rightarrow p(C = 1 | D = 0, T = 0) = \frac{p(C = 1, D = 0, T = 0)}{p(D = 0, T = 0)}$

Called *posterior probabilities*
or *conditional probabilities*

$$= \frac{0.008}{0.576 + 0.008} = 0.012$$

Sum rule

- Observe $D=1, T=1$?

$$p(C = 1 | D = 1, T = 1) = \frac{0.108}{0.016 + 0.108} = 0.871$$

T	D	C	P(T,D,C)
0	0	0	0.576
0	0	1	0.008
0	1	0	0.144
0	1	1	0.072
1	0	0	0.064
1	0	1	0.012
1	1	0	0.016
1	1	1	0.108

T	D	C	P(T,D,C)
0	0	0	0.576
0	0	1	0.008
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Posterior Conditional (FJDT) - Question

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<i>t</i>	<i>t</i>	<i>f</i>	<i>0.12</i>
<i>t</i>	<i>f</i>	<i>t</i>	<i>0.17</i>
<i>t</i>	<i>f</i>	<i>f</i>	<i>0.18</i>
<i>f</i>	<i>t</i>	<i>t</i>	<i>0.03</i>
<i>f</i>	<i>t</i>	<i>f</i>	<i>0.12</i>
<i>f</i>	<i>f</i>	<i>t</i>	<i>0.24</i>
<i>f</i>	<i>f</i>	<i>f</i>	<i>0.11</i>

4.e. (1 pt) $P(A = t \mid B = t) =$ _____

4.f. (1 pt) $P(C = f \mid B = t) =$ _____

Posterior Conditional (FJDT) - Solution

4. (10 pts total, 1 pt each) Probability Rules and Independence.

Consider the following full joint distribution for Boolean variables A, B, and C:

<i>A</i>	<i>B</i>	<i>C</i>	<i>P(a,b,c)</i>
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<i>t</i>	<i>t</i>	<i>f</i>	0.12
<i>t</i>	<i>f</i>	<i>t</i>	0.17
<i>t</i>	<i>f</i>	<i>f</i>	0.18
<i>f</i>	<i>t</i>	<i>t</i>	0.03
<i>f</i>	<i>t</i>	<i>f</i>	0.12
<i>f</i>	<i>f</i>	<i>t</i>	0.24
<i>f</i>	<i>f</i>	<i>f</i>	0.11

<i>A</i>	<i>B</i>	<i>C</i>	<i>P(a,b,c)</i>
<i>t</i>	<i>t</i>	<i>t</i>	0.03
<i>t</i>	<i>t</i>	<i>f</i>	0.12
<i>t</i>	<i>f</i>	<i>t</i>	0.17
<i>t</i>	<i>f</i>	<i>f</i>	0.18
<i>f</i>	<i>t</i>	<i>t</i>	0.03
<i>f</i>	<i>t</i>	<i>f</i>	0.12
<i>f</i>	<i>f</i>	<i>t</i>	0.24
<i>f</i>	<i>f</i>	<i>f</i>	0.11

$$\begin{aligned} P(A=t \mid B=t) &= P(A=t, B=t) \mid P(B=t) \\ &= (0.03 + 0.12) / (0.03 + 0.12 + 0.03 + 0.12) = 0.5 \end{aligned}$$

4.e. (1 pt) $P(A = t \mid B = t) =$ 0.5

4.f. (1 pt) $P(C = f \mid B = t) =$ 0.8

<i>A</i>	<i>B</i>	<i>C</i>	<i>P(a,b,c)</i>
<i>t</i>	<i>t</i>	<i>t</i>	0.03
<i>t</i>	<i>t</i>	<i>f</i>	0.12
<i>t</i>	<i>f</i>	<i>t</i>	0.17
<i>t</i>	<i>f</i>	<i>f</i>	0.18
<i>f</i>	<i>t</i>	<i>t</i>	0.03
<i>f</i>	<i>t</i>	<i>f</i>	0.12
<i>f</i>	<i>f</i>	<i>t</i>	0.24
<i>f</i>	<i>f</i>	<i>f</i>	0.11

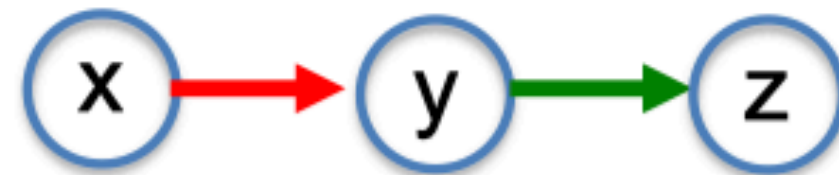
$$\begin{aligned} P(C=f \mid B=t) &= P(C=f, B=t) \mid P(B=t) \\ &= (0.12 + 0.12) / (0.03 + 0.12 + 0.03 + 0.12) = 0.80 \end{aligned}$$

Bayesian networks - Review

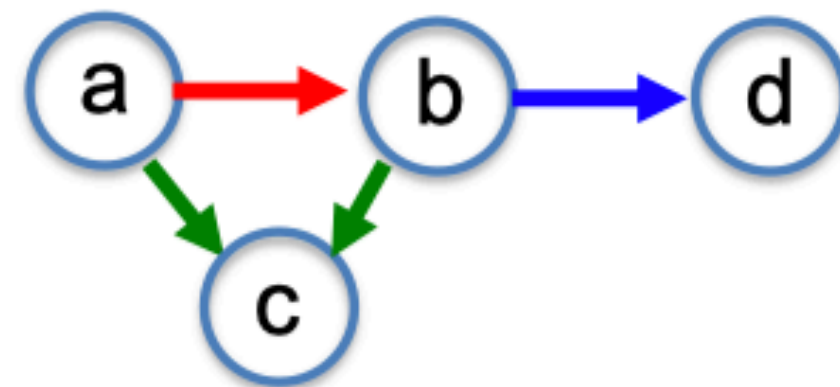
Bayesian networks

- Directed graphical model (DAG = directed acyclic graph)
- Nodes associated with variables
- “Draw” independence in conditional probability expansion
 - Parents in graph are the RHS of conditional

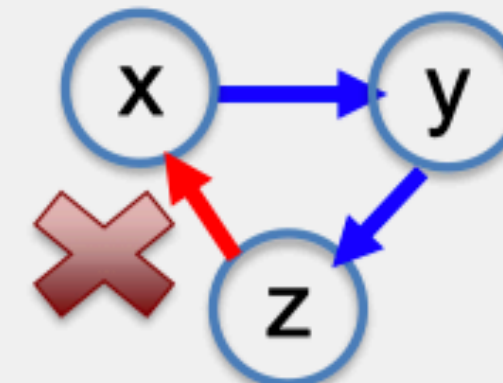
- Example: $p(x, y, z) = p(x) p(y | x) p(z | y)$



- Example: $p(a, b, c, d) = p(a) p(b | a) p(c | a, b) p(d | b)$



Graph must be **acyclic**

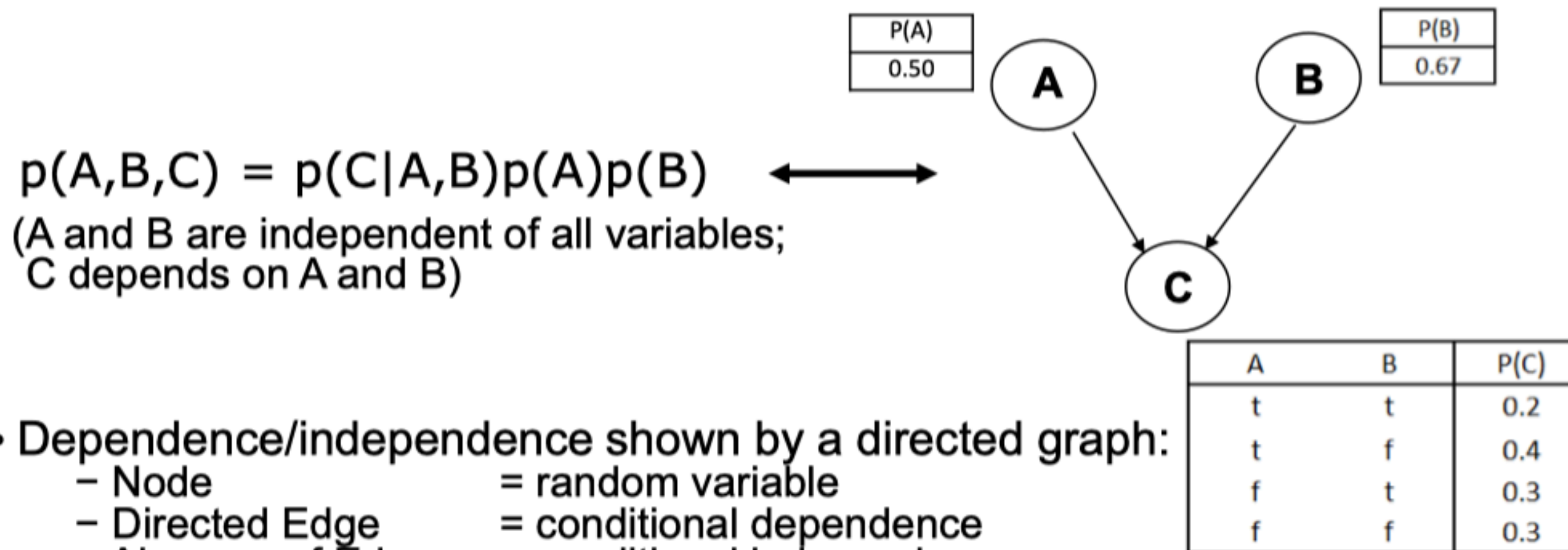


Corresponds to an order
over the variables
(chain rule)

Independence (BN) - Review

Bayesian network

- Specifies a joint distribution in a structured form:



- Dependence/independence shown by a directed graph:
 - Node = random variable
 - Directed Edge = conditional dependence
 - Absence of Edge = conditional independence
- Allows concise view of joint distribution relationships:
 - Graph nodes and edges show conditional relationships between variables.
 - Tables provide probability data.
- Tables are concise!!**
 - $P(\neg A)$ is not shown since it can be inferred as $(1 - P(A))$, etc.

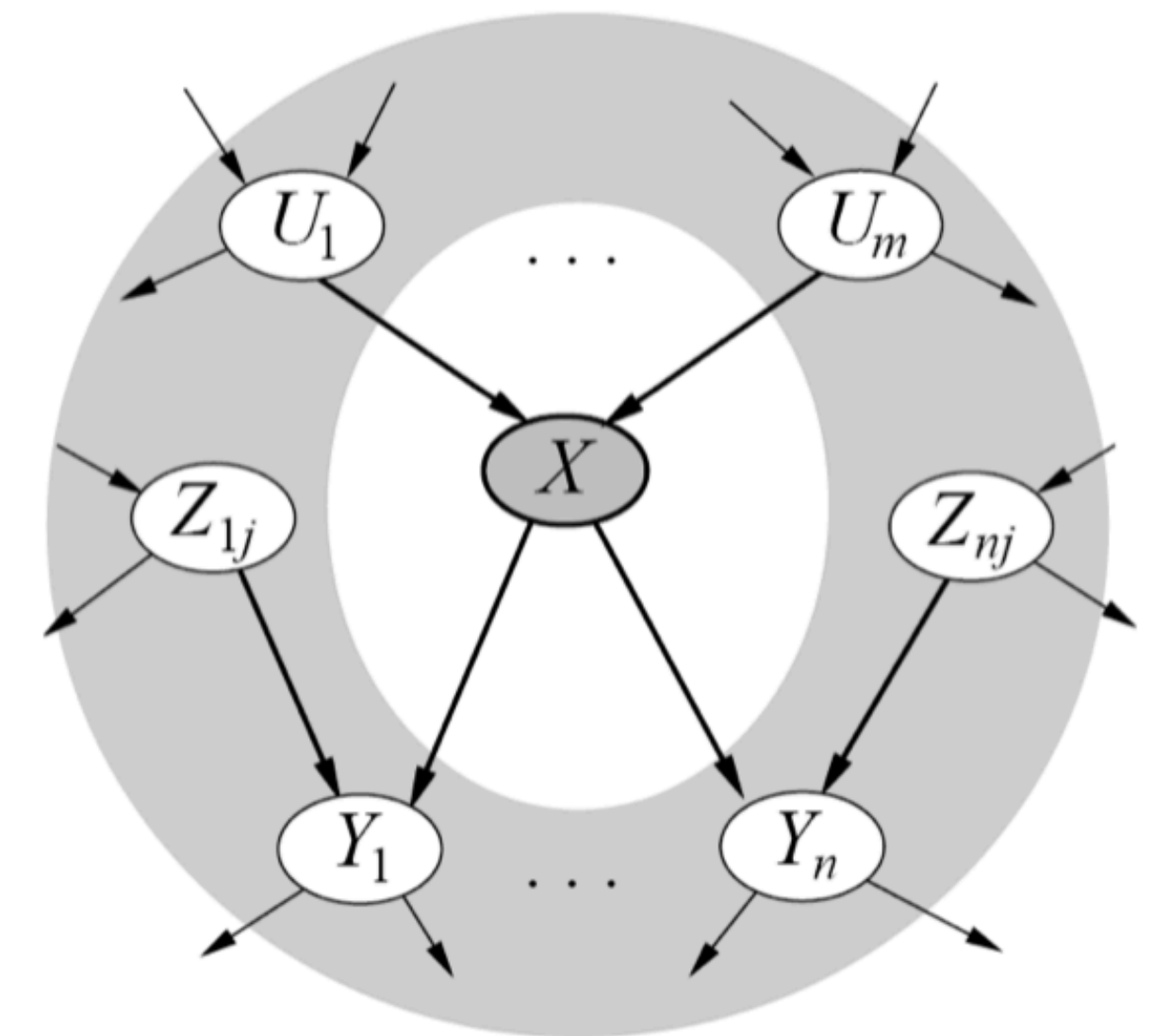
Given a graph, can we “read off” conditional independencies?

The “Markov Blanket” of X
(the gray area in the figure)

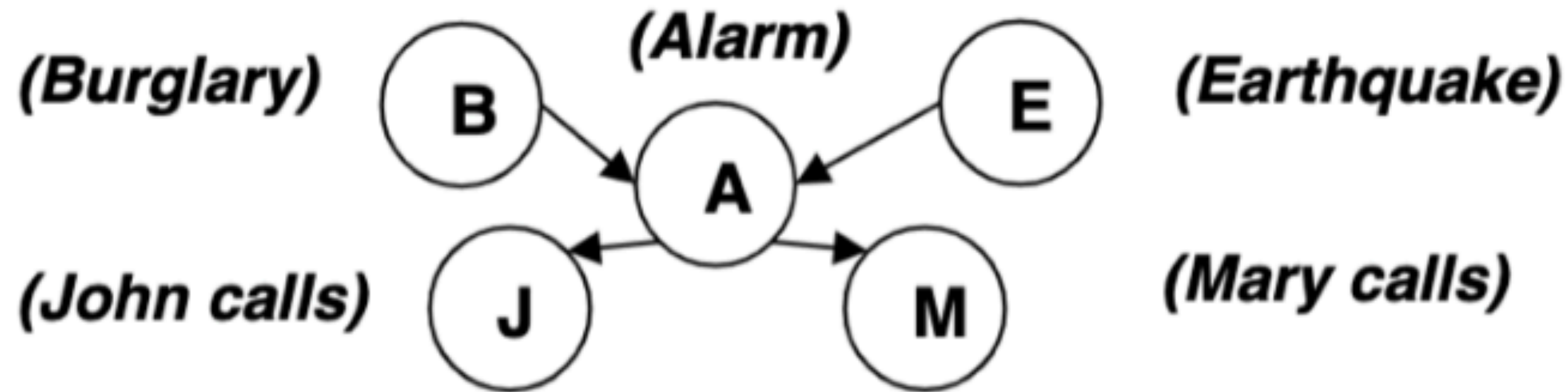
X is conditionally independent of everything else, GIVEN the values of:

- * X's parents
- * X's children
- * X's children's parents

X is conditionally independent of its non-descendants, GIVEN the values of its parents.

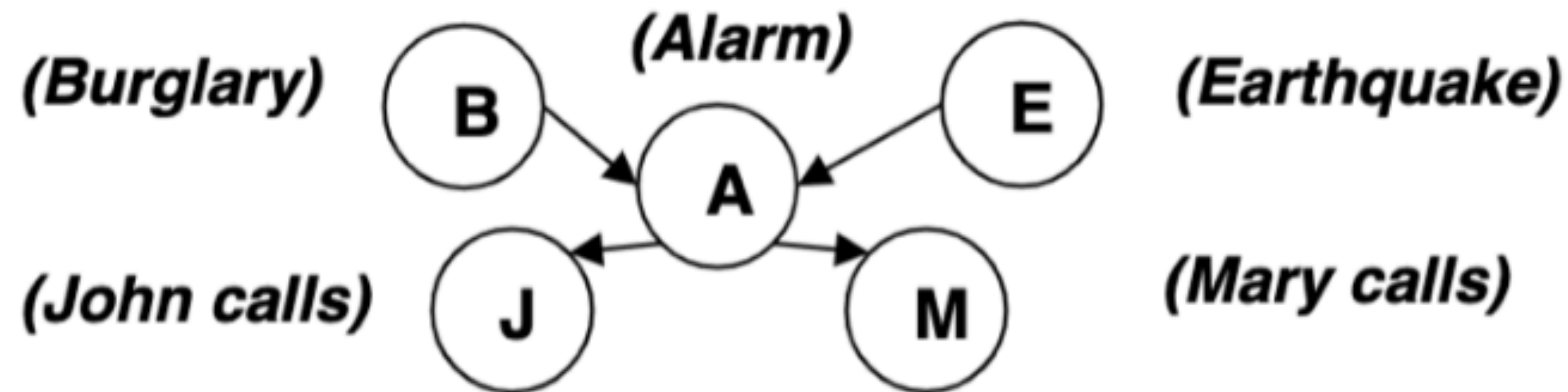


Independence (BN) - Question

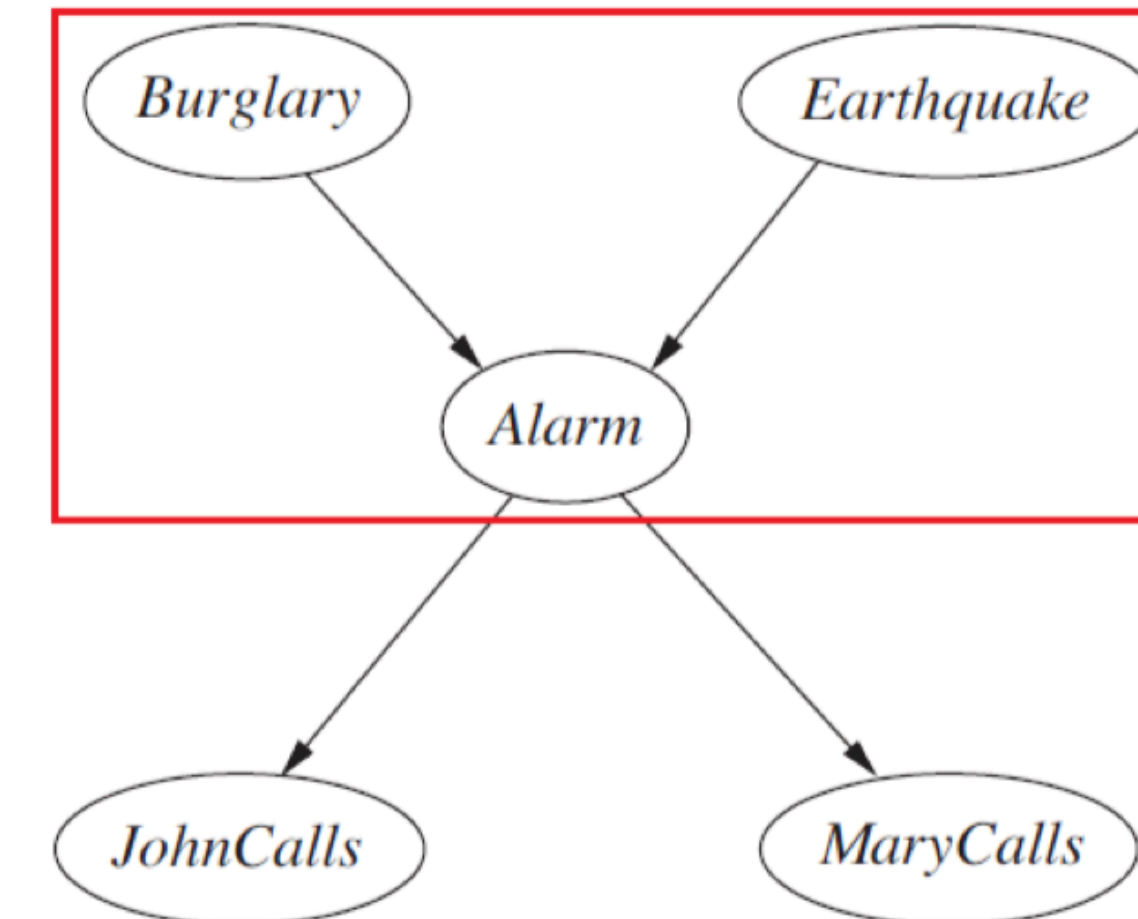


1. Are B and E independent of each other?
2. Are B and E independent of each other given A?
3. Are E and J independent of each other given A?
4. Are E and J independent of each other?
5. Are J and M independent of each other given A?
6. Are J and M independent of each other?

Independence (BN) - Solution

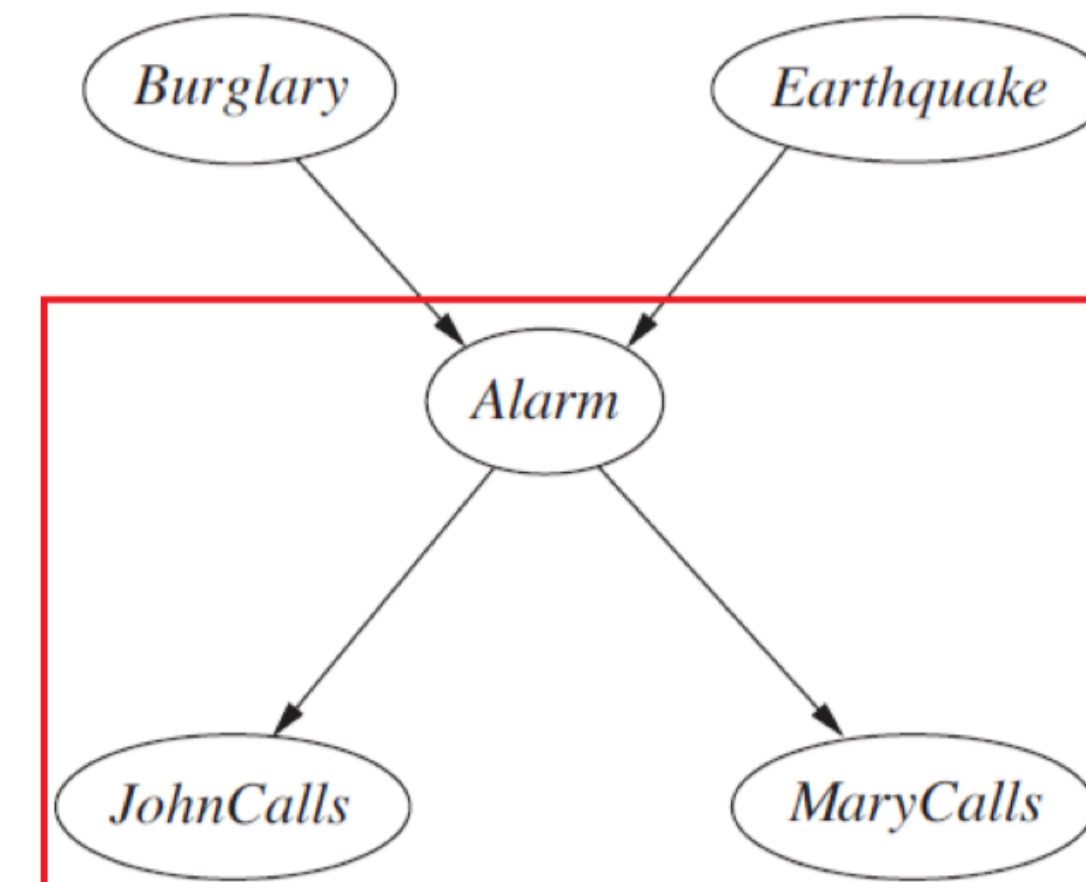


1. Are B and E independent of each other? **Yes**
2. Are B and E independent of each other given A? **No**
3. Are E and J independent of each other given A? **Yes**
4. Are E and J independent of each other? **No**
5. Are J and M independent of each other given A? **Yes**
6. Are J and M independent of each other? **No**



Assume Alarm not given -
are B and E independent? Yes.
No effect on each other.

Assume Alarm given -
are B and E independent? No.
If we check the news and no
reports of Earthquake, then
Burglary much more likely!
If Earthquake is reported, then
Burglary much less likely!
This is called "explaining away".



Assume Alarm not given -
are J and M independent? No.
If JohnCalls, then Alarm most
likely did go off, therefore
MaryCalls is probably true too!

Assume Alarm given -
are J and M independent? Yes.
MaryCalls (JohnCalls) is directly
influenced by Alarm, not the
other person. Once we know
Alarm, other person has no
effect.

Factoring (BN) - Review

Bayesian network

- Structure of the graph $\Leftrightarrow$ Conditional independence relations

In general,

$$p(X_1, X_2, \dots, X_N) = \prod p(X_i \mid \text{parents}(X_i))$$

The full joint distribution

The graph-structured representation

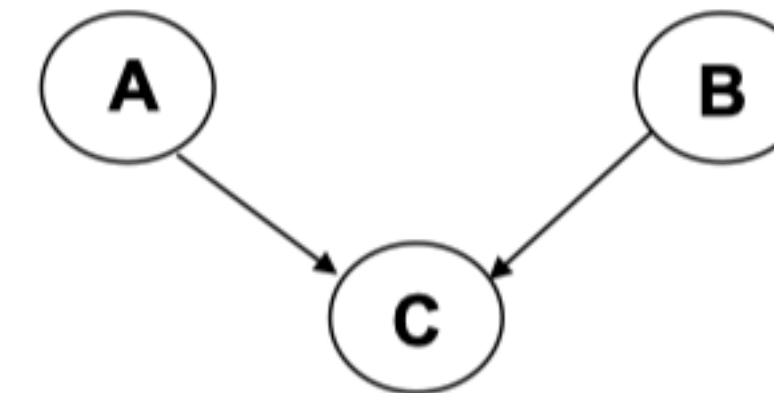
- Requires that graph is acyclic (no directed cycles)
- 2 components to a Bayesian network
 - The graph structure (conditional independence assumptions)
 - The numerical probabilities (for each variable given its parents)
- Also known as belief networks, causal networks
 - Example of graphical models
- Parents in the graph $\Leftrightarrow$ conditioning variables (RHS) in the formula

A, B, and C are independent.

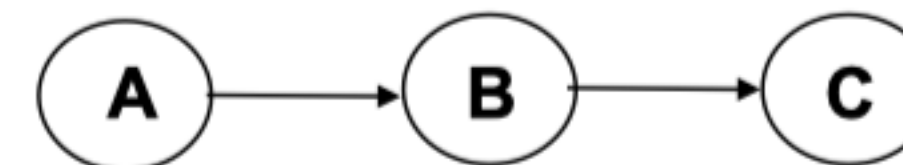


Marginal Independence:
 $p(A, B, C) = p(A) p(B) p(C)$

A and B directly influence C.



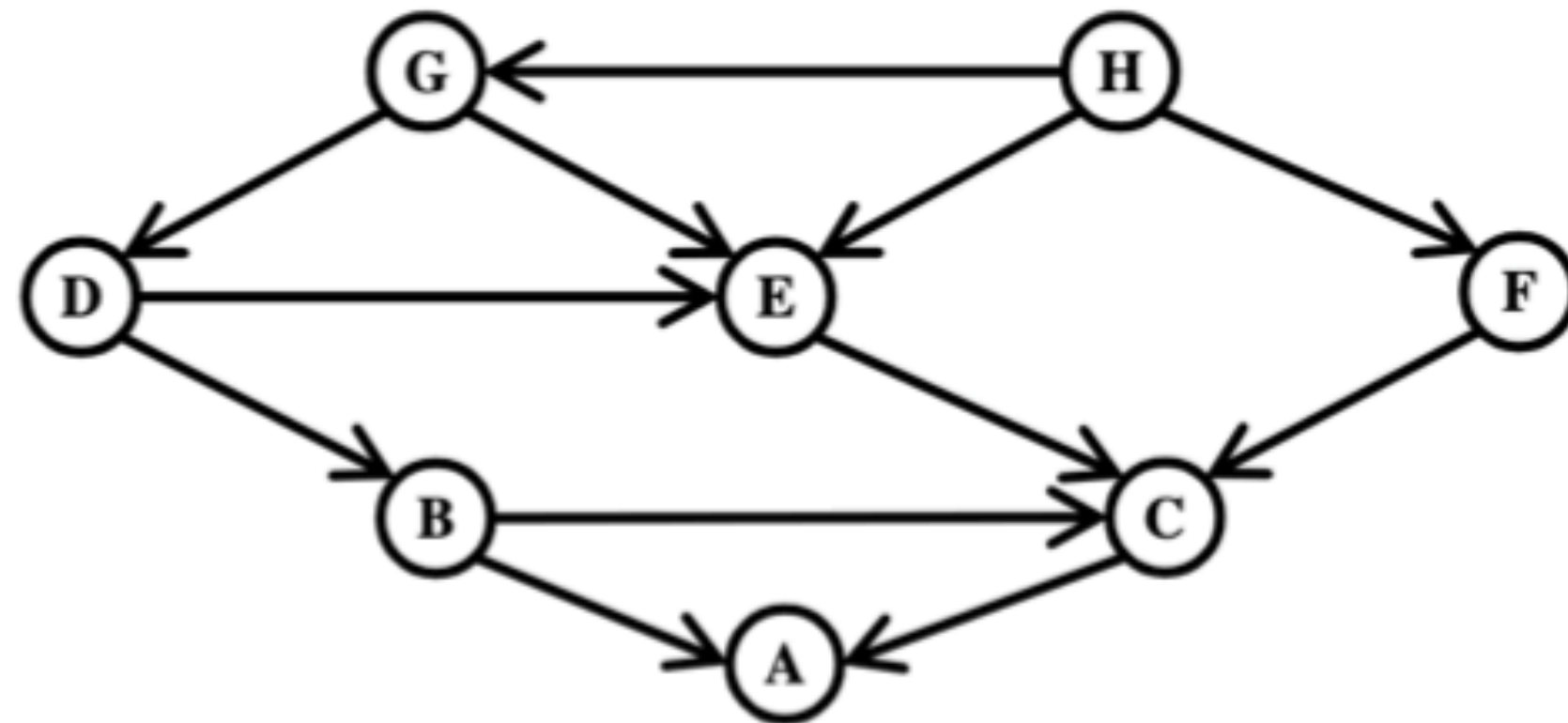
Independent Causes:
 $p(A, B, C) = p(C|A, B)p(A)p(B)$



Markov dependence:
 $p(A, B, C) = p(C|B) p(B|A)p(A)$

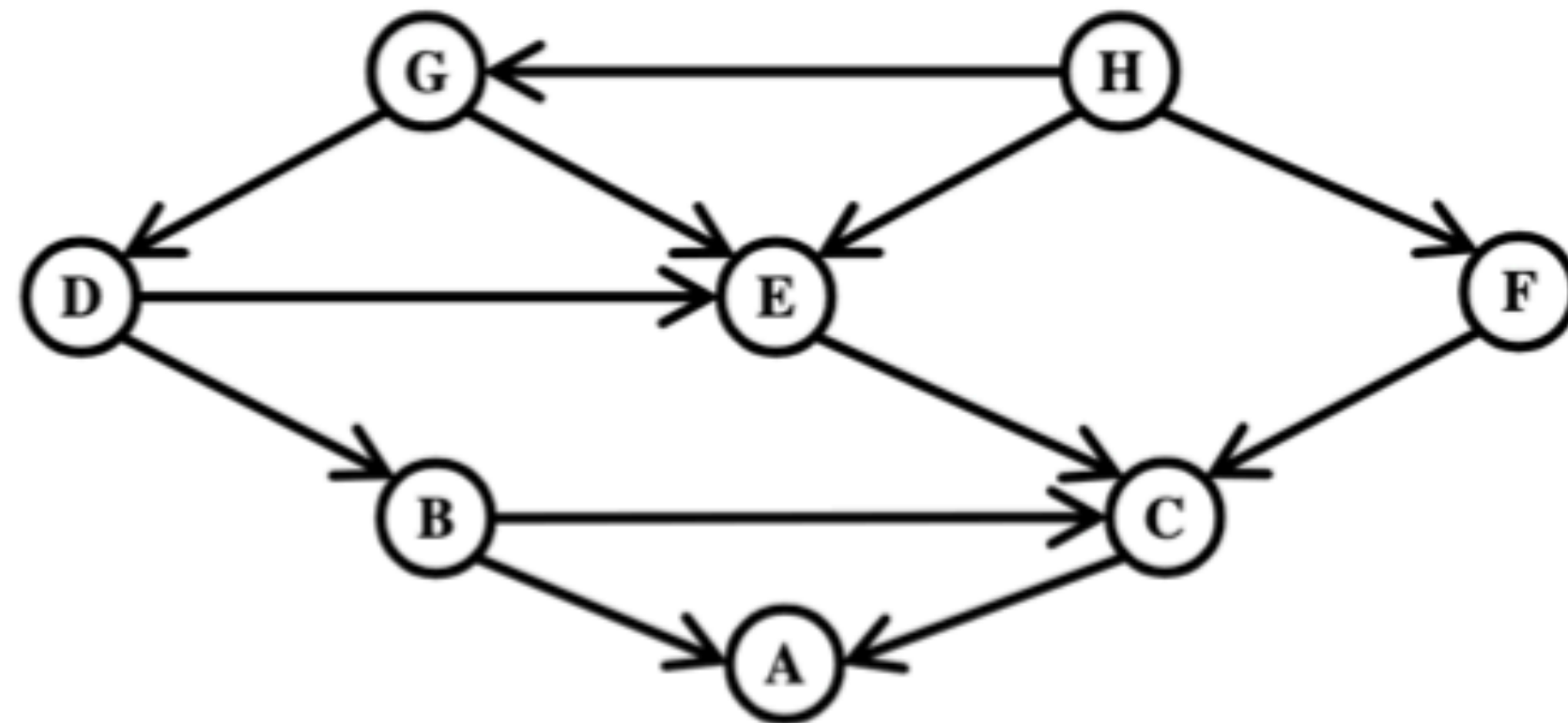
Factoring (BN) - Question

4b. (5 pts) Write down the factored conditional probability expression that corresponds to the graphical Bayesian Network shown.



Factoring (BN) - Solution

4b. (5 pts) Write down the factored conditional probability expression that corresponds to the graphical Bayesian Network shown.

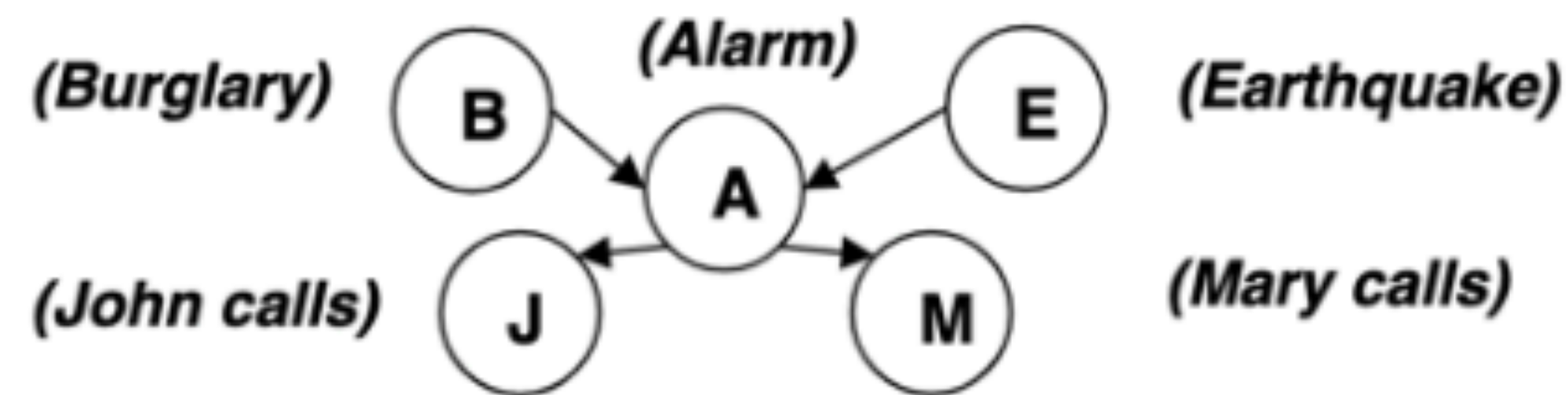


$$p(X_1, X_2, \dots, X_N) = \prod p(X_i \mid \text{parents}(X_i))$$

$$P(A \mid B, C) P(B \mid D) P(C \mid B, E, F) P(D \mid G) P(E \mid D, G, H) P(F \mid H) P(G \mid H) P(H)$$

Marginal probability (BN) - Question

4.c. (5 pts) Shown below is the Bayesian network corresponding to the Burglar Alarm problem,
 $P(J \mid A) P(M \mid A) P(A \mid B, E) P(B) P(E)$.



P(E)
.002

A	P(M)
t	.70
f	.01

P(B)
.001

A	P(J)
t	.90
f	.05

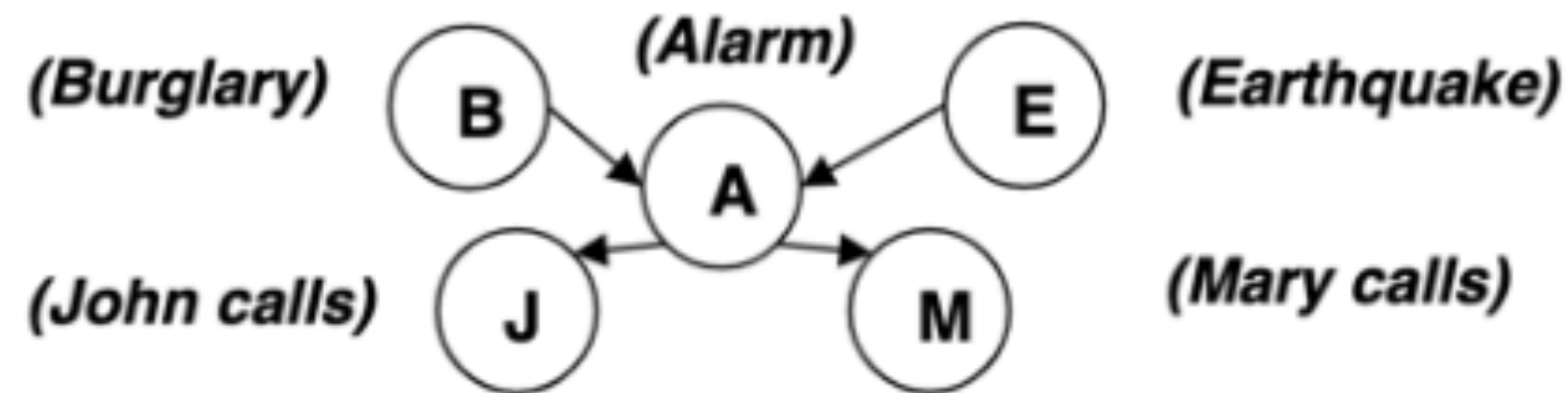
B	E	P(A)
t	t	.95
t	f	.94
f	t	.29
f	f	.001

The probability tables show the probability that variable is True, e.g., $P(M)$ means $P(M=t)$.
Compute the following probability:
 $P(j=t \wedge m=f \wedge a=f \wedge b=f \wedge e=t)$

Marginal probability (BN) - Solution

4.c. (5 pts) Shown below is the Bayesian network corresponding to the Burglar Alarm problem,

$P(J | A) P(M | A) P(A | B, E) P(B) P(E)$.



P(E)
.002

A	P(M)
t	.70
f	.01

P(B)
.001

A	P(J)
t	.90
f	.05

B	E	P(A)
t	t	.95
t	f	.94
f	t	.29
f	f	.001

The probability tables show the probability that variable is True, e.g., $P(M)$ means $P(M=t)$.

Compute the following probability:

$P(J=t \wedge M=f \wedge A=f \wedge B=f \wedge E=t)$

$$P(J, M, A, B, E) = P(J | A) * P(M | A) * P(A | B, E) * P(B) * P(E)$$

$$P(J=t, M=f, A=f, B=f, E=t)$$

$$= P(J=t | A=f) * P(M=f | A=f) * P(A=f | B=f, E=t) * P(B=f) * P(E=t)$$

$$= 0.05 * (1 - 0.01) * (1 - 0.29) * (1 - 0.001) * 0.002$$

$$= 7.02 * 10^{-5}$$

$$= 0.0000702$$